

Design Calculation Sheet VBF-T2R1SINGLEMODEL-CALC-01

Virtual Battery Factory — t2_r1_singlemodel — 2026-08-25

Parameter	Value	Unit	Source
Positive electrode thickness	100	um	params_r3.json
Negative electrode thickness	250	um	params_r3.json
Separator thickness	12.0	um	Chen2020
Electrode area	0.1027	m2	Chen2020 geometry
Negative exchange-current density	2.0	A/m2	params_r3.json
SEI kinetic rate constant	3.00e-13	m/s	params_r3.json
SEI reaction exchange current density	7.50e-08	A/m2	params_r3.json
EC diffusivity (SEI film)	1.0e-19	m2/s	params_r3.json (estimate)
1C discharge capacity	6.9555	Ah	cell/r3_1c_dfn.json
Discharge energy	25.03	Wh	cell/r3_energy.json
Gravimetric energy density	358.01	Wh/kg	cell/r3_energy.json
Volumetric energy density	625.0	Wh/L	cell/r3_energy.json
Cell mass (contract)	69.9	g	cell/r3_energy.json
N/P ratio	1.48	-	no capacity-density keys in Chen2020; full theoretical lithiation range (c_max*F/3600 cancel), c_max neg 33133 / pos 63104 mol/m3
Positive areal density	216.9	g/m2	process formula
Negative areal density	310.7	g/m2	process formula

Parameter	Value	Unit	Source
Positive compaction density	2.17	g/cm3	process formula
Negative compaction density	1.24	g/cm3	process formula
Electrolyte fill amount	0.0	g	pore volume x 1.2 g/cm3
SEI thickness @100 cyc	194.56	nm	cell/r3_aging100.json
SEI thickness @500 cyc	439.74	nm	cell/r3_aging500_d1e19.json
Low-T retention	99.59	%	cell/derived_retention_lowT_r3.json
4C plating ap_min	+0.112	V	cell/r3_4c45.json